

Indian Sign Language (ISL) Interpreter - User Manual

Complete Setup & Usage Guide for Developers and Users

1. Project Overview

This project is a real-time Indian Sign Language (ISL) Interpreter. It uses Google MediaPipe for 3D hand tracking and a custom Scale-Invariant Geometric Engine to accurately detect two-handed and single-handed signs regardless of camera distance, background, or lighting.

2. Prerequisites (For New Laptops)

Before running the project, ensure the following software is installed on the target machine:

- Python 3.8+ (IMPORTANT: Check the "Add Python to PATH" box during installation)
- Git (Optional, if you clone via terminal)

3. Step-by-Step Installation Guide

Open Command Prompt or PowerShell and run the following commands sequentially:

1. Clone or download the repository:
`git clone https://github.com/DENORYT/major-project.git`
2. Navigate into the folder:
`cd major-project`
3. Create a clean virtual environment:
`python -m venv venv`
4. Activate the virtual environment:
On Windows: `.\venv\Scripts\activate`
On Mac/Linux: `source venv/bin/activate`
5. Install required AI dependencies:
`pip install -r requirements.txt`

4. Running the Application

Once the dependencies are installed, you can start the application by running:

```
python Code/app.py
```

The terminal will output "Running on http://127.0.0.1:5000". Open your web browser (Chrome or Edge recommended) and navigate to that address to view the application.

Indian Sign Language (ISL) Interpreter - User Manual

Complete Setup & Usage Guide for Developers and Users

5. How to Use the Sentence Builder

The application features a real-time sentence builder that converts your sign language into fluid text. Here is how to use it:

- Typing Words: Hold a sign steady for ~1.2 seconds. The letter will automatically be appended to your sentence.
- Double Letters (e.g. "LL" in HELLO): Hold the sign for the first letter, drop your hand (or break the sign) for a split second, and then perform the sign again.
- Action Gestures:
 - * SPACE: Give a "Thumbs Up" to automatically add a space.
 - * BACKSPACE/DEL: Hold your "Pinky Finger Up" to delete the last letter.
- Speak: Click the "Speak" button to use Text-to-Speech to read your sentence out loud.

6. Complete Dictionary of Supported Signs

The custom mathematical engine tracks the exact 3D distances between your fingertips and palms to recognize the following 25 signs:

-- Single-Handed Signs --

0: Fist

1: Index finger up

2: Index and Middle fingers up

3: Thumb, Index, and Middle fingers up

4: Four fingers up

5: Open hand (all five fingers up)

C: Hand curved into a C shape

L: Thumb and Index finger extended in an L shape

-- Double-Handed ISL Signs --

6 to 9: 5 fingers on one hand (Base) + 1 to 4 fingers on the other hand (Pointer)

A: Pointer index tip touches Base thumb tip

B: Both hands open, touching side-by-side

D: Pointer index and thumb touch Base index (forming a D)

E: Pointer index tip touches Base index tip

I: Pointer index tip touches Base middle tip

M: Three fingers resting on Base palm

N: Two fingers resting on Base palm

O: Pointer index tip touches Base ring tip

S: Pinky fingers hooked together

Indian Sign Language (ISL) Interpreter - User Manual

Complete Setup & Usage Guide for Developers and Users

T: Pointer index finger resting on Base palm

U: Pointer index tip touches Base pinky tip

V: Two fingers in V shape resting on Base palm

W: Fingers interlocked (index to index, middle to middle)

X: Index fingers crossed